A392210

Odd-index evenness in a nested binomial equation

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AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

The normalized equation $D = x + (D(x/(1-x)))^2$ has a unique integral solution. Every odd-index coefficient beyond the first is even. We prove the source construction and the elementary characteristic-two square argument; the result formalizes an observed pattern, not a new general principle.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A392210 [1], retrieved on 24 October 2026. Its initial sequence index is 1. We also retain the formal zero constant coefficient at index zero.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x + O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U \circ V = U(V(x))$ only when $V(0) = 0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1 \rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

2 Formal algebra used in the proof

Lemma 2.1 (Triangular construction)**.** *Let K be a commutative ring. Suppose Φ preserves normalized series and improves agreement by one degree:*

$$G - H = O(x^r) \quad \Longrightarrow \quad \Phi(G) - \Phi(H) = O(x^{r+1}) \qquad (r \geq 2).$$

Then Φ has a unique normalized fixed point. The same conclusion holds for a family of series if the degree-n equations, at every family index, determine the degree-n coefficients from strictly lower degrees.

Proof. The degree-n coefficient of $\Phi(G)$ depends only on coefficients of G below degree n. Fix the constant and linear coefficients, and then define every later coefficient successively. This gives a fixed point; the same induction proves uniqueness. For a family, perform the induction on degree simultaneously at every family index. The same induction applies with only the constant coefficient prescribed, provided agreement is improved starting at degree one. It also applies to an operator on all series when agreement is improved starting at degree zero. For the

operators used in this paper, the recurrence requires only ring operations, so it also constructs integral solutions when $K = \mathbb{Z}$. Equivalently, iterating Φ from any normalized initial series fixes one further coefficient at each step. □

Lemma 2.2 (Inversion and transport)**.** *A normalized series over any commutative ring has a unique normalized two-sided compositional inverse. A constant-one series has a unique multiplicative inverse with constant coefficient one. These inverses and valid composition commute with coefficient ring homomorphisms. Normalized compositional inversion preserves agreement modulo x^r.*

Proof. For normalized G, the degree-n equation in $G(H) = x$ determines h_n with multiplier one from earlier coefficients. Construct a right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, so $H \circ G = x$ as well. The same associativity argument proves uniqueness. Multiplicative inversion follows similarly from the coefficient recurrence for $UV = 1$.

Each coefficient of a valid composition is a finite polynomial in the coefficients of its arguments. Ring homomorphisms preserve these polynomials and the defining equations, so uniqueness gives transport of inversion. If $G - H = O(x^r)$, composition with any normalized series preserves that agreement. Composing with the respective inverses then shows $\mathrm{Rev}(G) - \mathrm{Rev}(H) = O(x^r)$. □

3 A square-substitution construction

Lemma 3.1 (A useful square equation)**.** *For normalized $T, P \in K[[x]]$, there is a unique normalized solution of*

$$G = T + (G \circ P)^2.$$

The construction commutes with coefficient ring homomorphisms.

Proof. If $G - H = O(x^r)$, then $(G - H) \circ P = O(x^r)$. The sum $G \circ P + H \circ P$ has zero constant coefficient, so the difference of the two squares is $O(x^{r+1})$. Normalization is preserved. Lemma 2.1 gives existence and uniqueness; mapping the equation and using uniqueness gives transport. □

4 An elementary additional observation: A392210

The source gives the normalized functional equation [1]

$$D = x + \left(D\left(\frac{x}{1-x}\right)\right)^2. \tag{1}$$

It describes evenness at odd indices beyond the first as an apparent pattern. This follows directly from the square on the right.

Corollary 4.1. *Equation (1) has a unique normalized integral solution. Every odd-index coefficient beyond the first is even; equivalently, $[x^{2n-1}]D$ is even for every $n > 1$.*

Proof. The normalized series $M = x/(1-x)$ is defined by the actual unit inverse of $1 - x$. Lemma 3.1, with $T = x$ and $P = M$, constructs D and proves uniqueness for (1). Over $\mathbb{F}_2$, the square of any formal series has only even exponents. Therefore the only possibly nonzero odd-index coefficient on the right of (1) is the coefficient of x, which is one. All later odd-index coefficients of D vanish modulo two. □

5 Separate Lean proof and checked scope

The entry file is `lean/Entries/A392210.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A392210`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `third_existence`.
- `third_definition`.
- `third_evenness`.
- `third_arbitrary_source`.

The main implementation modules are

`Round4.NestedBinomial`

`Round4.SquareComposition`

The catalogue records the full original endpoint names and the new entry theorems. The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The collection report records source recompilation and transitive axiom checks. Accepted axioms are propositional extensionality, classical choice, and quotient soundness. All index conclusions above are unbounded; exact finite coefficient checks in the package provide separate corroboration.

Limits and status. This is the source apparent pattern and a direct characteristic-two square consequence. No analytic interpretation of the nested title is claimed. The manuscript consolidates completed results; it does not assert global novelty, exclusive priority, peer review, OEIS acceptance, or a new submission. Standard algebraic ingredients are not claimed as new discoveries.

References

[1] Paul D. Hanna, *OEIS A392210*, 2 February 2026. `https://oeis.org/A392210`. Retrieved 24 October 2026.